

Solution Problem 6.2

$$Y = Z\beta + \varepsilon$$

$$E[\varepsilon] = 0$$

$$\Sigma_e = \text{Cov}[\varepsilon] = E[\varepsilon\varepsilon^T] = \sigma^2 V$$

Multiply by $V^{-1/2}$

$$V^{-1/2}Y = (V^{-1/2}Z)\beta + V^{-1/2}\varepsilon$$

or

$$\boxed{Y^* = Z^*\beta + \varepsilon^*} \quad (*)$$

Now

$$\Sigma_{\varepsilon^*} = \text{Cov}[\varepsilon^*] = E[\varepsilon^* \varepsilon^{*T}]$$

$$= E[V^{-1/2} \varepsilon \varepsilon^T V^{-1/2}]$$

$$= V^{-1/2} E[\varepsilon \varepsilon^T] V^{-1/2}$$

$$= V^{-1/2} \sigma^2 V V^{-1/2}$$

$$= \sigma^2 I$$

$\Downarrow$ V symmetrical
 $V^{-1/2}$ symmetrical

$\Downarrow$ Thus (*) is on the form of standard OLS

$$\hat{\beta}_{OLS} = (Z^{*T} Z^*)^{-1} Z^{*T} Y^*$$

$$= (Z^T V^{-1/2} V^{-1/2} Z)^{-1} Z^T V^{-1/2} V^{-1/2} Y$$

$$= (Z^T V^{-1} Z)^{-1} Z^T V^{-1} Y$$

QED.